

Object Oriented Programming

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A bit about my self

- Metric @ FG school, HIT
- FSc @ Sir syed college, The mall, Rawalpindi
- BSc and MSc (Electronics & Communication)
 - @ UET Peshawar
- MSc (Embedded Systems) @ TU Delft
- PhD (Computer Engineering) @ TU Delft
- Post Doc (Quantum Computing) @ TU Delft

Research Interests

- Advanced Profiling (Memory and communication profiling)
- Accelerator-based computing (GPU/FPGA/DSP)
- Quantum Computing
- Compiler Technology
- Intelligent Systems



Netherlands/Holland



TU Delft



QCA Lab @ TU Delft



Quiz

- Who is:
 - Jan Koum
 - Brian Acton
- ?

Why Computers

Where do you find Computers?



Are computers dumb?

Why Programming

"I think everybody in this country should learn how to program a computer because it teaches you how to think."

- Steve Jobs

Why C++

Language Types (click to hide)

 Web  Mobile  Enterprise  Embedded

Language Rank	Types	Spectrum Ranking
1. Python	  	100.0
2. C++	  	98.3
3. C	  	98.3
4. Java	  	97.2
5. C#	  	92.7
6. R		82.8
7. PHP		82.7
8. JavaScript	 	82.6
9. Go	 	76.4
10. Assembly		74.2
11. Matlab		72.8
12. Scala	 	72.1
13. Ruby	 	71.3
14. HTML		70.5
15. Arduino		69.0
16. Shell		65.8
17. Perl	 	58.0
18. Swift	 	53.8
19. Processing	 	52.8
20. Objective-C	 	51.4
21. Fortran		51.1
22. Lua	 	49.3

Course Objectives

- To study and understand the object oriented programming paradigm.
- To justify the philosophy of object-oriented design and concepts of encapsulation, abstraction, inheritance and polymorphism
- To design, implement, test and debug simple and complex programming problems using object oriented programming.
- To design, implement and test the implementation of a “is-a” relationships among objects using a class hierarchy and inheritance

Course Objectives

- To compare and contrast the notion of overloading and overriding methods in an object-oriented language.
- To design, implement, test and debug event-driven programs that respond to user events.
- To develop code that responds to exception conditions raised during execution.

Course Organization

- 16 weeks (3+1 hr theory, 3 hr lab)
- 3 assignments
- 6 quizzes
- 2 sessional exams
- 1 final exam

Course Time Table

- Mon 1430 – 1630 hrs @ CSE 07
- Wed 1130 – 1230 hrs @ CSE 07

Course Grading

Theory (75 %)	Lab (25%)
Final Exam: 40%	Project
Sessional Exams: 25%	Performance
Assignments/Quizzes: 10%	Report

Course Books

- Object Oriented Programming in C++, 4 Edition, Robert Lafore
- C++ How to Program, 10th Edition, Deitel & Deitel

Review

- Computer?
- Computer programming?

Review

- Computer
 - Processor
 - Memory
 - Storage
- Computer programming: is a way of giving computers instructions (code) about what they should do next. Computer programmers write code to solve problems or perform a task.

Review (contd.)

- Algorithm?
- Computer Language?

Review (contd.)

- Algorithm is an unambiguous specification of how to solve a class of problems.
- Computer Language is a vocabulary and set of grammatical rules for instructing a computer or computing device to perform specific tasks.

Review (contd.)

- What are three basic ingredients of a programming language?

Review (contd.)

- Sequence
- Selection
- Repetition

Review (contd.)

- Constant, variable
- Data types
- If-else, loops
- Functions
- Structures
- Pointers

Reading for Next Week

- Functional Programming
- Structured Programming
- Imperative Programming
- Object Oriented Programming